

Problem: Representation of Experimental Data using Proper Orthogonal Decomposition (POD)

1 Introduction

The *Proper Orthogonal Decomposition* (POD) is a powerful method for analyzing and reducing the dimensionality of experimental or numerical data. It is widely used in fluid mechanics, signal processing, and data analysis. In this problem, we will apply POD to a set of simulated data representing velocity or temperature fields.

2 Provided Data

Each column of the matrix X represents a *snapshot* (an experimental or temporal realization) of a spatial field. For example, if X is of size $N \times M$, then:

- N is the number of spatial points,
- M is the number of *snapshots*.

3 Questions

3.1 Computing POD

POD relies on the singular value decomposition (SVD) or the diagonalization of the correlation matrix $C = \frac{1}{\alpha} X X^T$.

1. Compute the correlation matrix C from the matrix X .
2. Use MATLAB's `eig` function to compute the eigenvectors and eigenvalues of C .
3. What are the most significant eigenvalues?

3.2 Interpreting POD Modes

4. Plot the first five POD modes (eigenvectors associated with the three largest eigenvalues). Describe their spatial structure.
5. Compute the temporal coefficients associated with each POD mode by projecting the centered data onto the POD modes. Plot the temporal coefficients of the first five modes.

3.3 Data Reconstruction

6. Reconstruct the original data using only the first k POD modes, for $k = 1, 3, 5$. Calculate the relative reconstruction error for each case.
7. Discuss the influence of the number of modes on the reconstruction quality. What is the trade-off between dimensionality reduction and accuracy?

3.4 Application to Noisy Data

8. Add Gaussian noise to the original data (e.g., $X_{\text{noisy}} = X + 0.5 \times \text{randn}(N, M)$). Recompute the POD on this noisy data.
9. Compare the POD modes obtained with and without noise. Is POD robust to noise?